PHOSPHOGYPSUM IN M20 CONCRETE

Complete Material Testing & Concrete Testing Manual

A practical reference for laboratory records, project reports and presentations
Project focus: replacement of a portion of cement with phosphogypsum (PG), with M20 concrete as the base mix.

Prepared as a working guide. Confirm procedures, acceptance criteria and standard editions with your laboratory supervisor and the current BIS documents.

How to use this manual

This document collects common tests relevant to cement, fine aggregate, coarse aggregate, phosphogypsum and concrete. The procedures are concise study summaries, not substitutes for the complete Indian Standards (IS) or a laboratory SOP. Use the test methods actually performed in your project; do not report a test as completed unless it was carried out and recorded.

- Record sample source, date, test conditions, instrument identification and units.
- Use the same cube dimensions, curing conditions and loading procedure for all comparison groups.
- For PG replacement, state clearly whether replacement is by mass of cement or another basis. Do not mix replacement definitions.
- A 7-, 14- or 28-day result must be based on actual measured load and specimen dimensions; do not estimate missing values.
- Chemical, fluoride, phosphate, heavy-metal and radionuclide analyses should be done by a competent laboratory using validated methods.

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1. Cement tests

1.1 Specific gravity of cement

Aim: Determine the relative density of cement for mix-design calculations.

Apparatus/materials: Le Chatelier flask or suitable specific-gravity bottle, kerosene/naphtha as specified, balance, dry cement, temperature-controlled workspace.

Procedure (summary): Clean and dry the flask. Add the specified liquid and note the initial reading. Introduce a known mass of cement without trapping air, allow the liquid level to stabilize, and note the final reading. Follow the specified method for temperature and liquid correction.

Calculation/formula: Specific gravity = mass of cement / (displaced liquid volume × density of water); use the exact calculation in the applicable method.

Relevant standard: IS 4031 (Part 11) is commonly referenced for specific gravity of hydraulic cement; verify current edition. **|** **Reference:** Open test references

1.2 Fineness of cement

Aim: Assess particle fineness, which influences hydration and early strength.

Apparatus/materials: 75-micron IS sieve or air-permeability apparatus, balance, brush, timer.

Procedure (summary): For sieve method, weigh a representative dry sample, sieve for the prescribed duration, brush carefully and weigh residue. If using Blaine apparatus, follow the apparatus-specific procedure and calibration.

Calculation/formula: Sieve residue (%) = residue mass / initial sample mass × 100. Blaine method reports specific surface area as prescribed.

Relevant standard: IS 4031 (Part 1) for sieve fineness; IS 4031 (Part 2) for air-permeability fineness (confirm edition). **|** **Reference:** Open test references

1.3 Standard consistency

Aim: Determine the water content required to produce cement paste of standard consistency.

Apparatus/materials: Vicat apparatus with plunger, balance, measuring cylinder, cement, water, stopwatch.

Procedure (summary): Prepare paste using a trial water content and fill the Vicat mould. Release the plunger and observe penetration. Repeat with adjusted water until the penetration meets the standard-consistency criterion. Work rapidly and follow the prescribed timing.

Calculation/formula: Consistency (%) = water mass / cement mass × 100.

Relevant standard: IS 4031 (Part 4), Vicat method; verify current edition. | **Reference:** Open test references

1.4 Initial and final setting time

Aim: Determine the time from adding water until the paste reaches specified setting criteria.

Apparatus/materials: Vicat apparatus and needles, mould, balance, stopwatch, cement, water.

Procedure (summary): Start timing when water contacts cement. Prepare and fill the paste mould. Test at intervals using the specified needles and criteria; record initial and final setting times from the defined reference time.

Calculation/formula: Setting time = observed time meeting the specified needle-penetration criterion minus time of water addition.

Relevant standard: IS 4031 (Part 5), Vicat method; verify current edition. | **Reference:** Open test references

1.5 Soundness

Aim: Check expansion associated with potentially unsound cement.

Apparatus/materials: Le Chatelier apparatus, glass plates, water bath, measuring scale, balance.

Procedure (summary): Prepare cement paste, fill and condition the mould as specified, then measure the indicator separation before and after the prescribed heating/boiling procedure. Apply the standard's detailed timing and temperature requirements.

Calculation/formula: Expansion (mm) = final indicator separation − initial separation.

Relevant standard: IS 4031 (Part 3) is commonly used for soundness; verify current edition. | **Reference:** Open test references

2. Fine aggregate (sand) tests

2.1 Sieve analysis and grading

Aim: Determine particle-size distribution and grading zone of sand.

Apparatus/materials: Representative dry sand sample, IS sieves, sieve shaker, balance, trays, brush.

Procedure (summary): Oven-dry to constant mass if required. Weigh sample; arrange sieves in descending aperture size; shake for the prescribed period; weigh material retained on each sieve. Check total mass recovery.

Calculation/formula: Retained (%) = mass retained / total sample mass × 100. Cumulative retained (%) = running total. Passing (%) = 100 − cumulative retained (%). Fineness modulus = sum of cumulative percentages retained on the specified standard sieves / 100.

Relevant standard: IS 2386 (Part 1); grading limits should be checked against IS 383 and project specification. | **Reference:** Open test references

2.2 Specific gravity and water absorption

Aim: Determine relative density and absorption of fine aggregate.

Apparatus/materials: Pycnometer, balance, drying oven, trays, water, absorbent cloth; use the exact standard method.

Procedure (summary): Prepare the sample to the specified saturated-surface-dry (SSD) condition. Record masses required by the selected pycnometer method, dry the sample to constant mass, and calculate using the standard's defined symbols and equations.

Calculation/formula: Water absorption (%) = (SSD mass − oven-dry mass) / oven-dry mass × 100. Specific gravity depends on the defined oven-dry/SSD masses and pycnometer readings; use the standard equation.

Relevant standard: IS 2386 (Part 3); verify the method applicable to the aggregate. **|** **Reference:** Open test references

2.3 Silt / material finer than 75 µm

Aim: Assess excessive fines or clay/silt contamination in sand.

Apparatus/materials: Sieves or sedimentation jar as applicable, balance, water, measuring cylinder, drying oven.

Procedure (summary): Use the method specified by the project/lab. For sieve washing, wash a known dry mass over the prescribed fine sieve, dry retained material to constant mass and weigh. A jar test is only an indicative field check unless the specification accepts it.

Calculation/formula: Washed-out fines (%) = (initial dry mass − dry mass retained) / initial dry mass × 100.

Relevant standard: IS 2386 (Part 1) for material finer than the specified sieve; use project specification for acceptance. **|** **Reference:** Open test references

2.4 Bulking of sand (field/volume check)

Aim: Estimate volume increase in moist sand due to moisture films.

Apparatus/materials: Graduated cylinder or bulking container, water, sand, scale.

Procedure (summary): Measure the volume of moist sand in the container. Add water and stir to eliminate bulking; allow solids to settle and measure the reduced volume. Follow the lab method for exact readings.

Calculation/formula: Bulking (%) = (moist sand volume − inundated sand volume) / inundated sand volume × 100.

Relevant standard: Use the applicable laboratory/field procedure and project specification; not a substitute for grading or absorption tests. **|** **Reference:** Open test references

3. Coarse aggregate tests

3.1 Sieve analysis

Aim: Determine size distribution and nominal maximum size of coarse aggregate.

Apparatus/materials: Representative sample, IS sieves, balance, trays, sieve shaker, brush.

Procedure (summary): Dry and weigh the sample. Sieve through the selected sizes in sequence, weigh retained material and check mass recovery. Compare the grading with the project specification.

Calculation/formula: Retained (%) = retained mass / total mass × 100; passing (%) = 100 − cumulative retained (%).

Relevant standard: IS 2386 (Part 1); grading requirements commonly referenced from IS 383. **|** **Reference:** Open test references

3.2 Specific gravity and water absorption

Aim: Determine aggregate relative density and absorbed water.

Apparatus/materials: Wire basket or suitable container, water tank, balance, oven, trays, cloth.

Procedure (summary): Soak and bring the sample to SSD condition as specified. Measure SSD, submerged and oven-dry masses as required by the chosen method. Avoid surface water errors and record water temperature.

Calculation/formula: Water absorption (%) = (SSD mass − oven-dry mass) / oven-dry mass × 100. Calculate bulk/apparent specific gravity using the standard-defined mass terms.

Relevant standard: IS 2386 (Part 3); confirm the applicable method and current edition. **|** **Reference:** Open test references

3.3 Flakiness index

Aim: Determine the mass percentage of particles whose thickness is below the specified fraction of their mean size.

Apparatus/materials: IS sieves, thickness gauge, balance, trays.

Procedure (summary): Separate aggregate into standard size fractions. Test each fraction with the corresponding slot of the thickness gauge, weigh particles passing the gauge, and sum the masses. Follow the exact size-range and sampling requirements in the standard.

Calculation/formula: Flakiness index (%) = total mass of flaky particles / total mass of tested sample × 100.

Relevant standard: IS 2386 (Part 1); check current procedure and applicability. **|** **Reference:** Open test references

3.4 Elongation index

Aim: Determine the mass percentage of particles longer than the specified ratio for their size fraction.

Apparatus/materials: Length gauge, IS sieves, balance, trays.

Procedure (summary): Separate sample into specified size fractions. Pass particles through the matching length-gauge opening and weigh particles classified as elongated according to the standard. Record fractions and total mass.

Calculation/formula: Elongation index (%) = mass of elongated particles / total mass tested × 100.

Relevant standard: IS 2386 (Part 1); confirm the current method and size limits. **|** **Reference:** Open test references

3.5 Aggregate crushing value (ACV)

Aim: Assess resistance of aggregate to crushing under a prescribed compressive load.

Apparatus/materials: ACV cylinder, plunger, base plate, compression testing machine, sieves, balance.

Procedure (summary): Prepare aggregate in the prescribed size range and fill the cylinder in layers as specified. Apply the standard load/rate, sieve the crushed material through the prescribed sieve, and weigh the fines.

Calculation/formula: ACV (%) = mass passing the specified sieve after crushing / original sample mass × 100.

Relevant standard: IS 2386 (Part 4); follow the exact load and size range in the standard. **|** **Reference:** Open test references

3.6 Aggregate impact value (AIV)

Aim: Assess resistance to sudden impact.

Apparatus/materials: Impact testing machine, cup, hammer, tamping rod, sieves, balance.

Procedure (summary): Prepare the specified aggregate fraction. Fill and compact the cup according to the standard, apply the specified number of hammer blows, sieve the crushed material and weigh the fines.

Calculation/formula: AIV (%) = mass passing the specified sieve after impact / original sample mass × 100.

Relevant standard: IS 2386 (Part 4); follow the exact apparatus and blow sequence. **|** **Reference:** Open test references

4. Phosphogypsum characterization

PG composition varies with source and processing. Characterize the actual batch used in the concrete, and distinguish measured results from literature values. PG may contain soluble phosphate/fluoride, trace metals and naturally occurring radionuclides; use a competent laboratory and appropriate safety controls.

4.1 Physical appearance, moisture and sample preparation

Aim: Document the PG batch and prepare a representative, reproducible sample.

Apparatus/materials: Clean sample containers, balance, drying oven/desiccator where permitted, PPE, sample splitter.

Procedure (summary): Collect a representative composite sample from multiple points. Label source, date and processing state. Homogenize and reduce sample using a suitable splitter. Determine moisture only using a method validated for the sample and intended analysis.

Calculation/formula: Moisture (%) = (as-received mass − dry mass) / as-received mass × 100, when this definition matches the selected method.

Relevant standard: Use the laboratory's validated method; record drying temperature and duration because heating may alter the sample. **|** **Reference:** Open test references

4.2 Specific gravity / particle density

Aim: Determine particle density for material characterization and mix calculations where applicable.

Apparatus/materials: Pycnometer or suitable density apparatus, balance, liquid specified by the method, drying equipment.

Procedure (summary): Use a validated method compatible with PG solubility and particle characteristics. Record whether density is apparent, bulk or particle density, as these are not interchangeable.

Calculation/formula: Use the equation defined by the selected method; report units and test temperature.

Relevant standard: No single universal PG-specific IS method is assumed here; confirm a validated method with the testing laboratory. **|** **Reference:** Open test references

4.3 Particle-size distribution

Aim: Characterize PG fineness and particle-size distribution.

Apparatus/materials: Sieve stack or laser-diffraction analyzer, balance, sample splitter, dispersant if validated.

Procedure (summary): Choose dry sieving or laser diffraction based on fineness and agglomeration. Disperse consistently, run replicates where needed, and report the method and instrument settings.

Calculation/formula: For sieve analysis: retained (%) = mass retained / total mass × 100; passing (%) = 100 − cumulative retained (%). For laser diffraction, report D10, D50 and D90 as applicable.

Relevant standard: Use a validated particle-size method and report it; do not claim a specific IS standard unless the lab confirms it. **|** **Reference:** Open test references

4.4 XRF chemical composition

Aim: Determine major oxide composition of PG, such as CaO, SO3, SiO2 and P2O5.

Apparatus/materials: XRF spectrometer, prepared sample/pellet or fused bead, certified reference materials, trained operator.

Procedure (summary): A competent lab dries/prepares and homogenizes the sample, prepares a pressed pellet or fused bead according to the instrument method, calibrates with suitable standards and runs quality-control checks. Request the full report and analytical basis.

Calculation/formula: Report each oxide in mass percent as received or normalized; state loss-on-ignition and normalization basis. XRF oxide totals are not a direct safety assessment.

Relevant standard: Instrument/laboratory method; request the method, calibration and detection limits from the testing laboratory. **|** **Reference:** Open test references

4.5 Soluble phosphate and fluoride

Aim: Measure potentially leachable or soluble phosphate and fluoride relevant to PG use.

Apparatus/materials: Validated extraction apparatus, reagents, filtration equipment, ion chromatography or appropriate validated analytical instruments.

Procedure (summary): Use an extraction method appropriate to the intended exposure scenario and a validated analytical method. Include blanks, calibration, duplicates and reporting limits. Avoid improvised chemical procedures.

Calculation/formula: Report concentration with units and extraction ratio, such as mg/L in extract or mg/kg dry sample; do not compare unlike bases.

Relevant standard: Use an accredited laboratory method and applicable environmental/product requirements. **|** **Reference:** Open test references

4.6 Heavy metals and radionuclides (including Ra-226)

Aim: Evaluate source-specific contaminants and radiological characteristics where required.

Apparatus/materials: Accredited environmental/radiological laboratory; validated sample containers and analytical instruments.

Procedure (summary): Submit a representative sample to a competent lab. Request analytes, methods, detection limits, uncertainty and dry-mass basis. Follow applicable radiation-safety and waste-handling requirements.

Calculation/formula: Report analyte-specific results in the laboratory's stated units (for example, mg/kg or Bq/kg); do not infer safety from XRF alone.

Relevant standard: Applicable regulatory requirements and validated accredited-laboratory methods; consult local rules and supervisor. **|** **Reference:** Open test references

5. M20 mix design and fresh concrete

5.1 M20 mix design

Aim: Proportion materials to achieve the specified characteristic compressive strength and workability.

Apparatus/materials: Material test data, mix-design worksheet, balance, measuring containers, moisture data, trial-mix equipment.

Procedure (summary): Use the current IS 10262 method and relevant IS 456 provisions. Establish target mean strength using the prescribed standard deviation, choose water-cement ratio based on strength and durability, estimate water content and aggregate proportions, correct for moisture/absorption, then verify with trial mixes. For PG mixes, explicitly state the replacement basis and keep the control mix comparable.

Calculation/formula: Target mean strength = fck + 1.65s, where fck is characteristic strength and s is the applicable standard deviation. Final mix proportions must follow the standard and trial results.

Relevant standard: IS 10262 (concrete mix proportioning) and IS 456 (plain and reinforced concrete); verify latest revisions/amendments. **|** **Reference:** Open test references

5.2 Slump test (workability)

Aim: Measure consistency/workability of fresh concrete using the slump cone.

Apparatus/materials: Standard slump cone, base plate, tamping rod, ruler, scoop, fresh concrete.

Procedure (summary): Moisten apparatus as specified, fill cone in the prescribed layers, tamp each layer with the specified strokes, strike off the top, lift the cone vertically and measure the decrease in height promptly. Record slump type and value.

Calculation/formula: Slump (mm) = height of mould − height of slumped concrete at the specified measurement point.

Relevant standard: IS 1199 (fresh concrete testing; check applicable part and edition). **|** **Reference:** Open test references

5.3 Fresh concrete density (optional)

Aim: Determine fresh concrete mass per unit volume.

Apparatus/materials: Calibrated measure of known volume, balance, tamping/vibration equipment.

Procedure (summary): Fill and compact the measure using the method appropriate to the concrete. Strike level, clean the outside and weigh. Use the known calibrated volume.

Calculation/formula: Fresh density (kg/m³) = net concrete mass (kg) / measure volume (m³).

Relevant standard: Use applicable fresh-concrete test standard and laboratory SOP. **|** **Reference:** Open test references

6. Cube casting, curing and compressive strength

6.1 Cube mould preparation, casting and compaction

Aim: Prepare comparable specimens for compressive-strength testing.

Apparatus/materials: Clean cube moulds of specified dimensions (commonly 150 mm cubes), mould oil, weighing/mixing equipment, tamping rod or vibrator, labels.

Procedure (summary): Check mould dimensions and tightness. Apply a thin mould release layer. Mix the control and PG concrete using recorded batch quantities and mixing sequence. Fill and compact as specified, finish the top, label clearly and protect against moisture loss. Demould at the specified time and record any defects.

Calculation/formula: Record specimen dimensions and batch identification. Do not use a generic formula for casting; use the chosen standard's number of layers, compaction and curing requirements.

Relevant standard: IS 516 series for hardened concrete testing and relevant IS 1199 parts for sampling/fresh concrete; confirm current edition. **|** **Reference:** Open test references

6.2 Curing

Aim: Maintain specified moisture and temperature conditions until the test age.

Apparatus/materials: Curing tank or controlled curing room, clean water, thermometer, specimen labels.

Procedure (summary): After demoulding, place specimens in the prescribed curing environment. Keep labels legible, avoid damaging cube edges and record casting, demoulding and test dates. Use the same conditions for control and PG specimens.

Calculation/formula: Age at test = test date/time − casting date/time; report age in days.

Relevant standard: Relevant IS 516 / IS 456 provisions and laboratory SOP; confirm current requirements. **|** **Reference:** Open test references

6.3 Cube compressive strength (7, 14 and 28 days)

Aim: Measure compressive strength and compare control concrete with the selected PG replacement mix.

Apparatus/materials: Compression testing machine with current calibration, cube specimens, measuring scale, cleaning tools, test log.

Procedure (summary): At the specified age, remove the cube from curing, wipe surface water, measure dimensions if required, place it centrally in the machine with correct bearing faces, apply load continuously at the rate required by the current standard until failure, and record maximum load and failure pattern. Test the required number of specimens per age/mix.

Calculation/formula: Compressive strength (N/mm² or MPa) = maximum load (N) / loaded area (mm²). If load is in kN, convert to N by multiplying by 1000. For a 150 × 150 mm loaded face, area = 22,500 mm². Mean strength = sum of specimen strengths / number of specimens.

Relevant standard: IS 516 series for compressive strength of concrete; check the relevant current part/edition and machine loading-rate requirement. **|** **Reference:** Open test references

6.4 Strength comparison and percentage change

Aim: Compare PG concrete against control concrete at the same test age.

Apparatus/materials: Verified individual cube strengths for control and PG mixes at the same age.

Procedure (summary): Calculate each specimen's strength, then the mean for each mix/age. Compare only equivalent specimen sizes, curing conditions and test procedures. Report individual values as well as mean and spread.

Calculation/formula: Strength change (%) = (mean PG strength − mean control strength) / mean control strength × 100. Negative values indicate a reduction; positive values indicate an increase.

Relevant standard: Calculation based on measured results; report specimen count and variability. **|** **Reference:** Open test references

7. Optional durability / supplementary tests

Include these only if they were part of your approved methodology or you can perform them through the laboratory. They are not substitutes for compressive-strength results.

7.1 Water absorption of hardened concrete

Aim: Estimate water uptake by hardened concrete specimens under a defined conditioning method.

Apparatus/materials: Oven, balance, water tank, specimens, desiccator if specified.

Procedure (summary): Condition, dry, cool and weigh the specimen as required by the selected method. Immerse for the prescribed period, remove surface water consistently and weigh again. Do not mix different drying/immersion protocols.

Calculation/formula: Absorption (%) = (wet/SSD mass after immersion − dry mass) / dry mass × 100, where this matches the selected standard.

Relevant standard: Use the applicable standard/lab method and report exact conditioning durations and temperatures. **| Reference:** Open test references

7.2 Sorptivity / capillary water absorption

Aim: Assess capillary absorption under a controlled one-face exposure method.

Apparatus/materials: Conditioned concrete discs/cubes, balance, water tray, sealing material, timer.

Procedure (summary): Condition specimens using a validated method, seal sides if required, expose one face to a controlled water depth and record mass gain at prescribed intervals. Report exposed area and conditioning details.

Calculation/formula: Sorptivity is commonly obtained from the slope of cumulative absorption per exposed area versus square root of time; use the chosen standard's units and calculation.

Relevant standard: Use a validated published or laboratory method and cite it explicitly. **| Reference:** Open test references

7.3 Other durability tests (if required)

Aim: Assess permeability, chloride transport, sulphate resistance or other exposure-specific durability.

Apparatus/materials: Depends on test: permeability apparatus, chemical exposure setup, electrical test equipment, etc.

Procedure (summary): Select a test based on the project question and applicable standard. Use matched control and PG specimens, replicates and a predefined exposure duration.

Calculation/formula: Use the selected standard's reported metric; do not combine results from incompatible test methods.

Relevant standard: Specify the exact standard and edition selected by the supervisor/laboratory. **| Reference:** Open test references

8. Ready-to-fill result tables

8.1 Fine aggregate sieve analysis

Sieve size (mm)	Mass retained (g)	Retained (%)	Cumulative retained (%)	Passing (%)
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

8.2 Coarse aggregate sieve analysis

Sieve size (mm)	Mass retained (g)	Retained (%)	Cumulative retained (%)	Passing (%)
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

—	—	—	—	—

8.3 Material properties

Material	Test/property	Measured value	Unit / method / remarks
Cement	Specific gravity / fineness	—	—
Fine aggregate	Specific gravity / absorption	—	—
Coarse aggregate	Specific gravity / absorption	—	—
Coarse aggregate	Flakiness / elongation index	—	%
Phosphogypsum	Specific gravity / particle size	—	—
Phosphogypsum	XRF CaO	—	mass %
Phosphogypsum	XRF SO_3	—	mass %
Phosphogypsum	XRF P_2O_5	—	mass %
Phosphogypsum	Fluoride / radionuclide analysis	—	lab report units

8.4 Cube compressive-strength results

Mix ID	Age (days)	Cube ID	Size (mm)	Load (kN)	Area (mm²)	Strength (MPa)	Failure notes
Control	7	—	—	—	—	—	—
Control	14	—	—	—	—	—	—
Control	28	—	—	—	—	—	—
PG replacement	7	—	—	—	—	—	—
PG replacement	14	—	—	—	—	—	—
PG replacement	28	—	—	—	—	—	—

Strength formula: Strength (MPa) = Load (kN) × 1000 / loaded area (mm²). For 150 mm cubes, a 150 × 150 mm loaded face has an area of 22,500 mm². Confirm actual dimensions and the test standard.

8.5 Mean strength and comparison

Age	Control mean (MPa)	PG mix mean (MPa)	Change (%)	Remarks
7 days	—	—	—	—
14 days	—	—	—	—
28 days	—	—	—	—

8.6 Fresh concrete and mix design log

Batch ID	PG replacement basis/%	Water/binder ratio	Slump (mm)	Notes
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

9. Standards and consolidated reference links

These are clickable search/reference links to help locate the current standards and test explanations. Some standards may require purchase or institutional access. Verify exact titles, parts, revisions and amendments before citing.

- Bureau of Indian Standards (BIS)
- BIS standards search / Know Your Standard
- IS 4031 cement test methods
- IS 2386 aggregate test methods
- IS 383 coarse and fine aggregate specification
- IS 456 plain and reinforced concrete code
- IS 10262 concrete mix proportioning
- IS 516 concrete strength testing series
- IS 1199 fresh concrete test methods
- Cement test procedure references
- Fine aggregate test references
- Coarse aggregate test references
- Phosphogypsum characterization research
- Phosphogypsum in concrete research

Reporting checklist

- State the PG source, pretreatment/purification (if any), and replacement percentage/basis.
- Give all raw measurements, units, equations, mean values and specimen counts.
- Include photographs only for tests actually performed and label them accurately.
- Identify the applicable IS code, part, edition and amendments confirmed by the laboratory.
- Do not claim a mix meets M20 characteristic strength from one cube or one age alone; follow the applicable acceptance criteria.
- Discuss limitations, repeatability, variability and any safety/environmental test results separately.

End of manual. This is a study and record-keeping aid; current standards and the laboratory supervisor take precedence.